Stochastic Finance and Risk Modelling

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4 - Optimal portfolios

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Outline

Reward and risk

Market portfolio $w^0 = 0$

General portfolio

Program

Reward and risk

Market portfolio $w^0 = 0$

General portfolio

$\mathbb{P}$ vs $\mathbb{Q}$: from seller to buyer

The market consists of 1 riskless asset S_t^0 and d risky assets $S_t = (S_t^1, \ldots, S_t^d)$ on some probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

- We focus on the one-period model, i.e. $T = 1$.
- Sellers aim to build a "fair" pricing framework for trading financial products — arbitrage-free condition and risk-neutral $\mathbb{Q} \sim \mathbb{P}$.
- In what follows we stand for buyers, and our goal is to optimize "utility function" — maximise reward and minimize risk.

Portfolio

- A portfolio, denoted by $V^{\xi^0,\xi}$, is identified by the trading strategy $(\xi^0, \xi) \in \mathbb{R} \times \mathbb{R}^d$, i.e.

$$V_t^{\xi^0,\xi} := \xi^0 S_t^0 + \xi \cdot S_t, \quad t = 0, 1.$$

- Set

$$w^i := \frac{\xi^i S_0^i}{V_0^{\xi^0,\xi}} = \frac{\xi^i S_0^i}{\sum_{i=0}^d \xi^i S_0^i} \quad \text{for } i = 0, \dots, d.$$

- Clearly, $\sum_{i=0}^{d} w^i = 1$. If $w^i > 0$ (resp. $w^i < 0$) means long (resp. short) position of S^i. If $w^i \geq 0$, w^i denotes the proportion of $V_0^{\xi^0,\xi}$ invested in the ith asset.

$$V_t^{\xi^0,\xi} = V_0^{\xi^0,\xi} \sum_{i=0}^{d} w^i \frac{S_t^i}{S_0^i}, \quad t = 0, 1.$$

- If $w := (w^1, \dots, w^d) : w^0 + w \cdot \mathbf{1} = 1, \quad$ with $\mathbf{1} := (1, \dots, 1) \in \mathbb{R}^d$.

Reward and risk

- Recall that the return of $V^{\xi^0,\xi}$ is defined if $V_0^{\xi^0,\xi} \neq 0$ by

$$\begin{aligned}
\rho(V^{\xi^0,\xi}) &:= \frac{V_1^{\xi^0,\xi} - V_0^{\xi^0,\xi}}{V_0^{\xi^0,\xi}} = \frac{\xi^0(S_1^0 - S_0^0) + \xi \cdot (S_1 - S_0)}{V_0^{\xi^0,\xi}} \\
&= \frac{\xi^0 S_0^0}{V_0^{\xi^0,\xi}} \frac{S_1^0 - S_0^0}{S_0^0} + \sum_{i=1}^{d} \frac{\xi_i S_0^i}{V_0^{\xi^0,\xi}} \frac{S_1^i - S_0^i}{S_0^i} \\
&= w^0 \rho(S^0) + \sum_{i=1}^{d} w^i \rho(S^i) =: w^0 \rho(S^0) + w \cdot \rho(S).
\end{aligned}$$

  Recall that $\rho(S^0) = r$ where $S_t^0 = S_0^0(1+r)^t$ and $\rho(S) := (\rho(S^1), \ldots, \rho(S^d))$.

- As $w^0 = 1 - w \cdot \mathbf{1}$, from now we write V^w instead of $V^{\xi^0,\xi}$.

Reward and risk

- The expected return of the portfolio V^w is given

$$r(V^w) := \mathbb{E}^{\mathbb{P}}[\rho(V^w)] \equiv \mathbb{E}[\rho(V^w)] = (1 - w \cdot \mathbf{1})R^0 + w \cdot R$$

where $R^i = \mathbb{E}[\rho(S^i)] = \mathbb{E}\left[\frac{S_1^i}{S_0^i}\right] - 1$ for $i = 0, \ldots, d$.

- We assume that $R = (R_1, \ldots, R_d)$ is not colinear to $\mathbf{1}$.
- The risk of the portfolio V^w is given

$$\begin{aligned} Var(\rho(V^w)) &:= \mathbb{E}\big[(\rho(V^w) - \mathbb{E}[\rho(V^w)])^2\big] \\ &= \sum_{1 \le i,j \le d} w^i w^j Cov(\rho(S_t^i), \rho(S_t^j)) = w^T \Sigma w, \end{aligned}$$

where $\Sigma := \big(\sigma_{i,j}\big)_{1 \le i,j \le d}$ and

$$\sigma_{i,j} := Cov(\rho(S_t^i), \rho(S_t^j)) = \frac{Cov(S_t^i, S_t^j)}{S_0^i S_0^j}.$$

- We assume that Σ is invertible and positive def, i.e. $w \neq 0 \Rightarrow w^T \Sigma w > 0$.

Objective

Let $\sigma > 0$ and $\mu \in \mathbb{R}$.

- Maximize the reward $\mathbb{E}[\rho(V^w)]$ under the risk constraint :

$$\max_{w \in E} \big((1 - w \cdot \mathbf{1})R^0 + w \cdot R\big) \quad \text{s.t.} \quad w^T \Sigma w \leq \sigma^2.$$

- Minimize the variance $Var(\rho(V^w))$ under a reward constraint :

$$\min_{w \in E} w^T \Sigma w \quad \text{s.t.} \quad (1 - w \cdot \mathbf{1})R^0 + w \cdot R \geq \mu.$$

- Optimize

$$\max_{w \in E} \left((1 - w \cdot \mathbf{1})R^0 + w \cdot R - \frac{\lambda}{2} w^T \Sigma w \right)$$

The subset $E \subset \mathbb{R}^d$ stands for trading constraints for the risky assets and $\lambda \geq 0$ denotes the risk aversion parameter.

Program

Reward and risk

Market portfolio $w^0 = 0$

General portfolio

Market portfolio : $w^0 = 0$ or $w \cdot \mathbf{1} = 1$

We start by considering $w^0 = 0$ and set $E = E_0 := \{w \in \mathbb{R}^d : w \cdot \mathbf{1} = 1\}$. Then one obtains respectively

$$f(\sigma) := \max_{w \in E_0} w \cdot R \quad \text{s.t.} \quad w^T \Sigma w \leq \sigma^2 \tag{1}$$

$$g(\mu) := \min_{w \in E_0} w^T \Sigma w \quad \text{s.t.} \quad w \cdot R \geq \mu \tag{2}$$

$$\max_{w \in E_0} \left(w \cdot R - \frac{\lambda}{2} w^T \Sigma w \right). \tag{3}$$

To illustrate, take $d = 3$ and

$$R = \begin{pmatrix} 0.018 \\ 0.025 \\ 0.01 \end{pmatrix} \quad \text{and} \quad \Sigma = \begin{pmatrix} 1.08 & 0.34 & 0.38 \\ 0.34 & 3.09 & -1.59 \\ 0.38 & -1.59 & 1.54 \end{pmatrix}.$$

Example

Solving (2) for $\mu = 0.018$,

- one obtains the optimizer $\tilde{w} = (0.046, 0.509, 0.445)$;
- one has $\tilde{w} \cdot R = 0.018 = \mu$ and $g(\mu) = \tilde{w}^T \Sigma \tilde{w} = 0.422$;
- Another feasible $w = (1, 0, 0) \in E_0$ satisfying $w \cdot R = \mu$ yields the variance $\sigma_{1,1} = w^T \Sigma w = 1.08 > g(\mu)$;
- If comparing the two portfolios $\tilde{w}$ and w, we notice that the variance of the return of $\tilde{w}$ is about four times below $\sigma_{1,1}$. This means that $\tilde{w}$ is much more diversified.

Solving similarly (1) for $\sigma^2 = \sigma_{1,1}$,

- one obtains the optimizer $\tilde{w} = (0.282, 0.69, 0.028)$;
- one has $\tilde{w}^T \Sigma \tilde{w} = \sigma^2$ and $f(\sigma) = \tilde{w} \cdot R = 0.0226$;
- $w = (1, 0, 0) \in E_0$ is still feasible as $w^T \Sigma w = \sigma^2$, while the corresponding expected return is $R_1 = 0.018 < f(\sigma)$.

Mean-variance efficient frontier

We continue the analysis by describing the set of all optimal portfolios.

- The function $[\underline{R}, \overline{R}] \ni \mu \to g(\mu) \in \mathbb{R}_+$ is well defined by solving the corresponding minimization problems (2), where $\underline{R} := \min_{1 \leq i \leq d} R^i$ and $\overline{R} := \max_{1 \leq i \leq d} R^i$.
- We can determine the trade-off, known as the efficient frontier, between variance σ and expected return μ. The efficient frontier can also be obtained by solving (1). The difference is that we vary the upper bound on the variance and maximize the expected return.
- By varying the risk aversion parameter λ and solving (3), we obtain the mean-variance efficient portfolios.
- If $\lambda = 0$, then we obtain the portfolio with maximum expected return.
- If $\lambda \gg 1$, then the relative importance of the variance in the objective function becomes much greater.

Lagrange method without the condition $w \cdot \mathbf{1} = 1$

To solve

$$\min_{w \in \mathbb{R}^d} w^T \Sigma w \quad \text{s.t.} \quad \mathbf{w} \cdot R \geq \mu.$$

We introduce the Lagrange multiplier

$$L(w, \lambda) := w^T \Sigma w + \lambda(\mu - w \cdot R).$$

It yields (first order conditions)

$$\mathbf{0} = 2\Sigma w - \lambda R \quad \text{and} \quad w \cdot R - \mu = 0$$

and

$$\tilde{w} = \mu \frac{\Sigma^{-1} R}{R^T \Sigma^{-1} R}.$$

Lagrange method

To solve

$$\min_{w \in E_0} w^T \Sigma w \quad \text{s.t.} \quad \mathbf{w} \cdot R \geq \mu,$$

recall that $E_0 := \{w \in \mathbb{R}^d : w \cdot \mathbf{1} = 1\}$. We introduce the Lagrange multipliers

$$L(w, \lambda, \theta) := w^T \Sigma w + \lambda(\mu - w \cdot R) + \theta(1 - w \cdot \mathbf{1}).$$

It yields

$$\mathbf{0} = 2\Sigma w - \lambda R - \theta \mathbf{1} \quad \text{and} \quad w \cdot R - \mu = 0 = w \cdot \mathbf{1} - 1$$

and

$$\tilde{\lambda} = 2\frac{\mu c - b}{ac - b^2}, \quad \tilde{\theta} = 2\frac{a - \mu b}{ac - b^2}, \quad \tilde{w} = \frac{1}{2}\tilde{\lambda}\Sigma^{-1}R + \frac{1}{2}\tilde{\theta}\Sigma^{-1}\mathbf{1},$$

where

$$a := R^T \Sigma^{-1} R, \quad b := R^T \Sigma^{-1} \mathbf{1}, \quad c := \mathbf{1}^T \Sigma^{-1} \mathbf{1}.$$

Efficient frontier

Recall that

$$\tilde{\lambda} = 2\frac{\mu c - b}{ac - b^2}, \quad \tilde{\theta} = 2\frac{a - \mu b}{ac - b^2}, \quad \tilde{w} = \frac{1}{2}\tilde{\lambda}\Sigma^{-1}R + \frac{1}{2}\tilde{\theta}\Sigma^{-1}\mathbf{1},$$

$$\begin{aligned}
\tilde{w} &= \frac{\mu c - b}{ac - b^2}\Sigma^{-1}R + \frac{a - \mu b}{ac - b^2}\Sigma^{-1}\mathbf{1} \\
&= \mu\frac{c\Sigma^{-1}R - b\Sigma^{-1}\mathbf{1}}{ac - b^2} + \frac{a\Sigma^{-1}\mathbf{1} - b\Sigma^{-1}R}{ac - b^2} \\
&= \mu h + g,
\end{aligned}$$

where

$$h = \frac{c\Sigma^{-1}R - b\Sigma^{-1}\mathbf{1}}{ac - b^2} \text{ and } g = \frac{a\Sigma^{-1}\mathbf{1} - b\Sigma^{-1}R}{ac - b^2}.$$

Efficient frontier

- Every portfolio of the efficient frontier can be written as $h\mu + g$.
- g is the risky asset allocation corresponding to $\mu = 0$, i.e. a zero return.
- $g + h$ is the risky asset allocation corresponding to $\mu = 1$.
- Theorem of separation into two funds : in this model, it is enough to invest in two funds instead than in the d risky assets.

Illustration

Take the previous example, we plot $\sigma \mapsto f(\sigma)$:

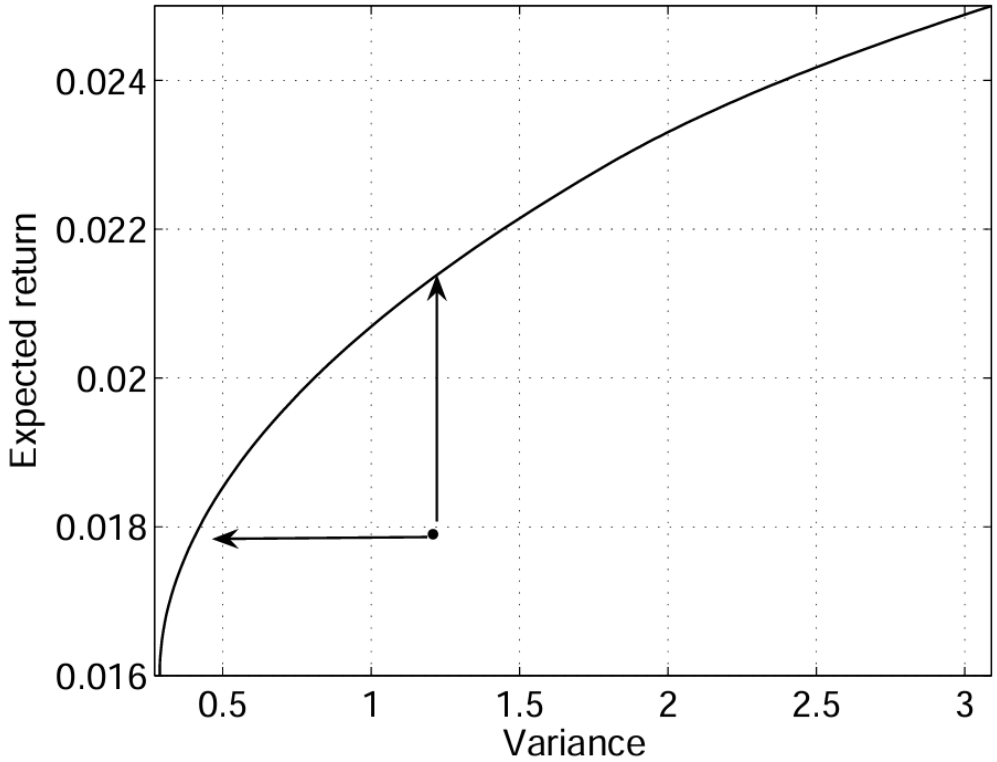

Figure – The curve is the efficient frontier. The dot indicates the position of a sub-optimal portfolio and the arrows indicate the position of the optimal portfolios obtained by minimizing variance or maximizing expected return.

Program

Reward and risk

Market portfolio $w^0 = 0$

General portfolio

Riskless asset added

As a result, the minimization problem (2) becomes

$$\min_{(w^0,w)\in\mathbb{R}\times E} w^T\Sigma w \quad \text{s.t.} \quad w^0R^0 + w\cdot R \geq \mu \text{ and } w^0 + w\cdot\mathbf{1} = 1$$

and the equivalent problems (1) and (3) change accordingly.

- The new set of mean-variance efficient portfolios is obtained by varying the lower bound μ on the expected return ;
- The optimal portfolio is always a combination of one portfolio of the risky assets and the riskless asset ;
- Changing the lower bound μ results in different relative proportions of the two.

An alternative point of view

- For each $w^0 \in I := (-\infty, 1)$, one has $\hat{w} \cdot \mathbf{1} = 1$ with $\hat{w} := w/(1-w^0)$ and $\hat{\mu}(w^0) := (\mu - w^0 R^0)/(1-w^0)$

$$\min_{(w^0,\hat{w}) \in I \times E_0} (1-w^0)^2 \hat{w}^T \Sigma \hat{w} \text{ s.t. } \hat{w} \cdot R \geq \hat{\mu}(w^0)$$

$$\Longrightarrow \min_{w^0 \in I} (1-w^0)^2 g(\hat{\mu}(w^0)).$$

- All efficient portfolios can be represented as

$$\rho(V) = \tilde{w}^0 R^0 + (1-\tilde{w}^0)\tilde{w} \cdot \rho(S),$$

  where $(1-\tilde{w}^0)\tilde{w}$ denotes the scaled weights of the market portfolio and $\tilde{w}^0$ is the weight of the riskless asset ;
- The market portfolio is located on the efficient frontier, where a straight line passing through the location of the riskless asset is tangent to the efficient frontier. The straight line is knownasthe capital market line and the market portfolio is also known as the tangency portfolio.

Illustration

Figure below shows the efficient frontier of the previous example but with standard deviation instead of variance :

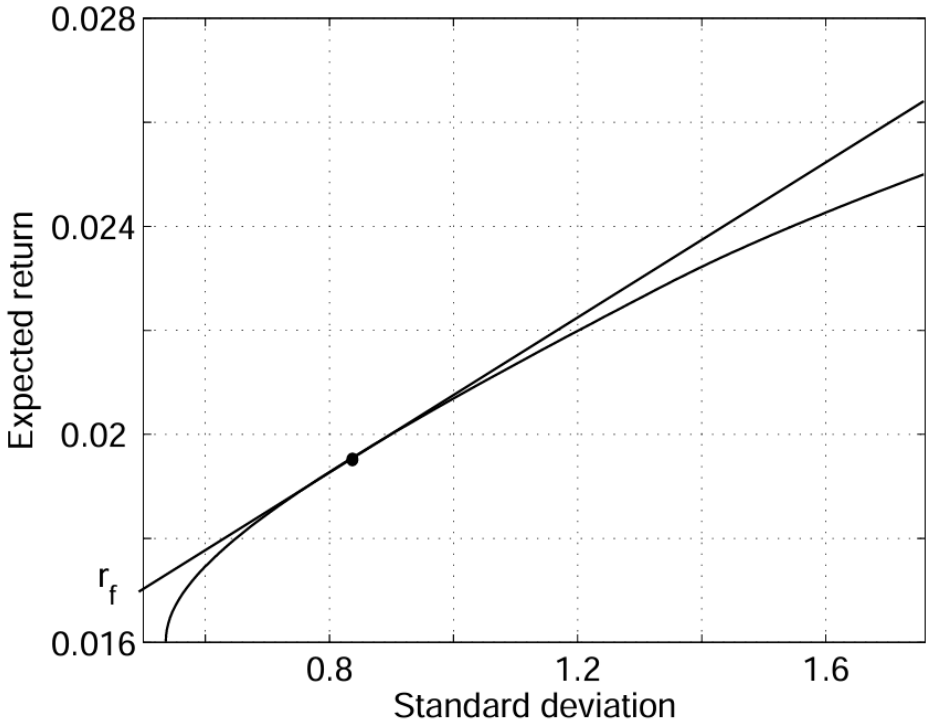

Figure – The dot indicates the position of the market portfolio, where the capital market line is tangent to the efficient frontier.The interest rate r is shown on the vertical axis and the straight line is the capital market line.

Equation of the capital market line

Rewrite $V_M := \tilde{w} \cdot S$. Then

$$\rho(V) = \tilde{w}^0 \rho(S^0) + (1 - \tilde{w}^0)\rho(V_M).$$

- Taking the expectation on both sides,

$$r(V) = \tilde{w}^0 R^0 + (1 - \tilde{w}^0) r(V_M) = R^0 + (1 - \tilde{w}^0)\big(r(V_M) - R^0\big);$$

- Taking the variance on both sides,

$$Var(\rho(V)) = (1 - \tilde{w}^0)^2 Var(\rho(V_M));$$

- We derive the capital market line equation

$$r(V) = R^0 + \sqrt{\frac{Var(\rho(V))}{Var(\rho(V_M))}}\big(r(V_M) - R^0\big),$$

which describes the efficient frontier with the riskless asset added to the investment universe.